Kooshan Amini

Machine Learning Research Scientist

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Ph.D. expected Fall 2027 · Available for Summer 2027 internships · No visa sponsorship required

SUMMARY

Ph.D. candidate at Rice University developing computer vision, foundation-model, and probabilistic methods for remote sensing and risk. I adapt vision-language and depth foundation models to scarce, shifting real-world imagery, evaluate open VLMs at scale, and couple learned models with physics constraints, Bayesian fusion, and Monte Carlo uncertainty. Five journal papers (four first-author), 130+ citations, and field citation ratios up to 18.8.

RESEARCH INTERESTS

Multimodal and vision-language models · remote sensing and geospatial AI · generalization under distribution shift · uncertainty quantification · physics-informed learning · data-efficient fine-tuning and evaluation

RESEARCH EXPERIENCE

Graduate Research Assistant, Rice University — Aug 2022 – Present

Co-advised by Engineering and Computer Science faculty; projects funded by NSF, NASA, NIST, and Harris County

Vision-language segmentation under domain shift — NSF

- Co-led CLIPSeg-debris: fine-tuned only the decoder of a frozen CLIP-based model (1.2M of 150M parameters) with visual-prompt engineering and text–image prompt mixing (co-first author, CACAIE 2025).
- Designed a cross-storm split and multi-annotator consensus labels: Dice 0.86 / IoU 0.83 on an unseen hurricane vs. 0.54 zero-shot, ahead of DeepLabV3 (0.82) and U-Net (0.79); SAM and Grounded-SAM fail zero-shot.
- Reported a useful negative result: single-region training beat pooled multi-region data on 3 of 4 unseen regions.

Monocular 3D estimation from single aerial images — NSF

- Built DebrisHeightNet: a 1.1M-parameter head on frozen Depth Anything V2 (ViT-L) conditioned on segmentation masks, trained on LiDAR–monocular fusion targets with a Gaussian-likelihood loss and an under-prediction penalty.
- Matched UAV surveys with Spearman $\rho = 0.87$ across 10 regions and 5 hurricanes; region-level calibration cut the median error factor from 2.7× to 1.9×.
- Single-image estimates fall within 30% of hauled volume, where Hazus and FEMA methods over-predict 4.8× and 2.7×.

Zero-shot vision-language model evaluation on satellite imagery — NASA ROSES

- Benchmarked 4 open VLMs on road and bridge damage: reasoning mode, set-of-mark prompting, and FFT co-registration raised damage recall from 0.33 to 0.78; identified output-token truncation of reasoning.
- Designed a 1,000-item benchmark with exact inclusion probabilities, spatially blocked splits, and blind dual screening; ran inference on 30,362 segments in 3.1 days.

Physics-informed learning and uncertainty quantification — NSF

- Built a monotonicity-constrained network (18 increasing and 2 decreasing features) after GPs, gradient boosting, and unconstrained DNNs produced implausible extrapolations.
- Coupled it with sea-level-rise and land-cover scenarios in 9,000 Monte Carlo runs: debris +11–14% and emergency-access loss +9–17% by 2040 (ASCE Natural Hazards Review, in press).
- Random-field debris dispersion with fragility models and network analysis (13,410 road segments) showed debris raises loss of emergency-medical access from 2% to 17% (RESS 2023; field citation ratio 14.8).

Bayesian data fusion for flood situational awareness — NSF OpenSafe.AI

- Fused crowd reports, traffic cameras, and gauges with 11 forecast cycles of 160 scenarios: held-out AUC from 0.654 to 0.707 in 11 of 11 cycles; flagged 738 of 1,286 reports a median 27 h early.
- Designed the multi-team system architecture and its interface standards.

Decision science for community resilience — NIST Center of Excellence

- Multi-objective optimization over an integrated hazard–damage–recovery–economics model (80 scenarios) found a $200M portfolio that cuts expected losses by $192M; efficiency did not guarantee equity (first of 16 authors).

Risk & Resilience Engineering Intern, Arup — Jun – Aug 2025

- Delivered flood-risk and transportation-resilience analyses for industry and public-sector clients; full-time offer.

Research Assistant & Software Developer, Sharif University of Technology — 2018 – 2022

- Probabilistic models of earthquake-induced social impacts; developer on Rtx, a reliability and risk platform.

PUBLICATIONS

[1] **Amini, K.**, Meads, M. M., Padgett, J. E., et al. (2026). Evolving threats: Understanding the coupled impact of sea level and land-cover changes on hurricane-induced debris risk. *ASCE Natural Hazards Review*, in press.

[2] **Amini, K.**, Padgett, J. E., & Balakrishnan, G. (2026). Rapid debris-volume estimation from post-hurricane aerial imagery. *arXiv:2608.17165*; under review at *ISPRS Journal of Photogrammetry and Remote Sensing*.

[3] **Amini, K.**, Palit, T., Beck, A. L., et al. (2026). Integrated decision-support framework for enhancing coastal community resilience against hurricane-induced hazards. *International Journal of Disaster Risk Reduction*, 140, 106136.

[4] **Amini, K.***, Liu, Y.*, Padgett, J. E., Balakrishnan, G., & Veeraraghavan, A. (2025). Debris segmentation using post-hurricane aerial imagery. *Computer-Aided Civil and Infrastructure Engineering*, 40(25), 4116–4131. (*equal contribution)

[5] **Amini, K.**, & Padgett, J. E. (2023). Probabilistic risk assessment of hurricane-induced debris impacts on coastal transportation infrastructure. *Reliability Engineering & System Safety*, 240, 109579.

[6] Nofal, O. M., **Amini, K.**, Padgett, J. E., van de Lindt, J. W., et al. (2023). Multi-hazard socio-physical resilience assessment of hurricane-induced hazards on coastal communities. *Resilient Cities and Structures*, 2(2), 67–81.

Also: conference papers at ICOSSAR 2025 and ICASP14 2023. 130+ citations; field citation ratios 14.8 and 18.8 (Dimensions).

OPEN SCIENCE

- Released CLIPSeg-debris weights (Hugging Face), a 1,242-image consensus-labeled dataset, and research code; the DesignSafe datasets reached 778 views and 246 downloads.
- Merged an electric power network recovery analysis (+1,417 lines) into NCSA's open-source IN-CORE platform.

TALKS, AWARDS & SERVICE

- **Talks:** invited webinars for NDPTC (2026), NSF NHERI DesignSafe (2025), and NHERI SimCenter (2024, the only graduate-student speaker); conference talks at AEI/IRD, ICOSSAR, ASCE EMI, ICASP14, and NHERI.
- **Awards:** selected for the 1st IN-CORE Contributor Cohort (NCSA and NIST, 2024); NHERI travel award (2023); national exams: 58th of 39,392 (top 0.2%) in graduate engineering, 16th in the engineering Scientific Olympiad.
- **Service:** journal reviewer for *Climatic Change* and *Natural Hazards and Earth System Sciences*; expert novelty assessor for the UK Metascience Novelty Indicators Challenge (UKRI).
- **Teaching & leadership:** D2K Research Mentoring Fellow (2026); teaching assistant for a graduate resilience course (2025); Vice President, Graduate Student Association (2022–2024).

RESEARCH SKILLS

Methods	foundation-model fine-tuning, segmentation, monocular depth, VLM benchmarking, Bayesian inference, Gaussian processes, monotonic networks, random fields, Monte Carlo, ALE/PDP
Frameworks	PyTorch, PyTorch Lightning, Hugging Face Transformers, vLLM, scikit-learn, OpenCV, Weights & Biases
Data & tools	Python (expert), SQL, C++, MATLAB, GeoPandas, rasterio/GDAL, LiDAR/DEM, OpenStreetMap, NetworkX, CuPy, Git, Linux/HPC (TACC), Label Studio

EDUCATION

Ph.D. in Engineering, Rice University · GPA 3.96/4.00 — 2022 – Expected Fall 2027

M.Sc. in Engineering, Sharif University of Technology · GPA 3.42/4.00 — 2017 – 2020

B.Sc. in Engineering, University of Guilan · GPA 3.88/4.00, ranked 3rd of 80+ — 2013 – 2017